RTS3917N

3 Mega Pixel H.265/H.264 IP Camera SoC

Brief DATASHEET

Rev 1.0
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USING THIS DOCUMENT

This document is intended for the hardware and software engineer's general information on the Realtek RTS3917N IP camera SoC.

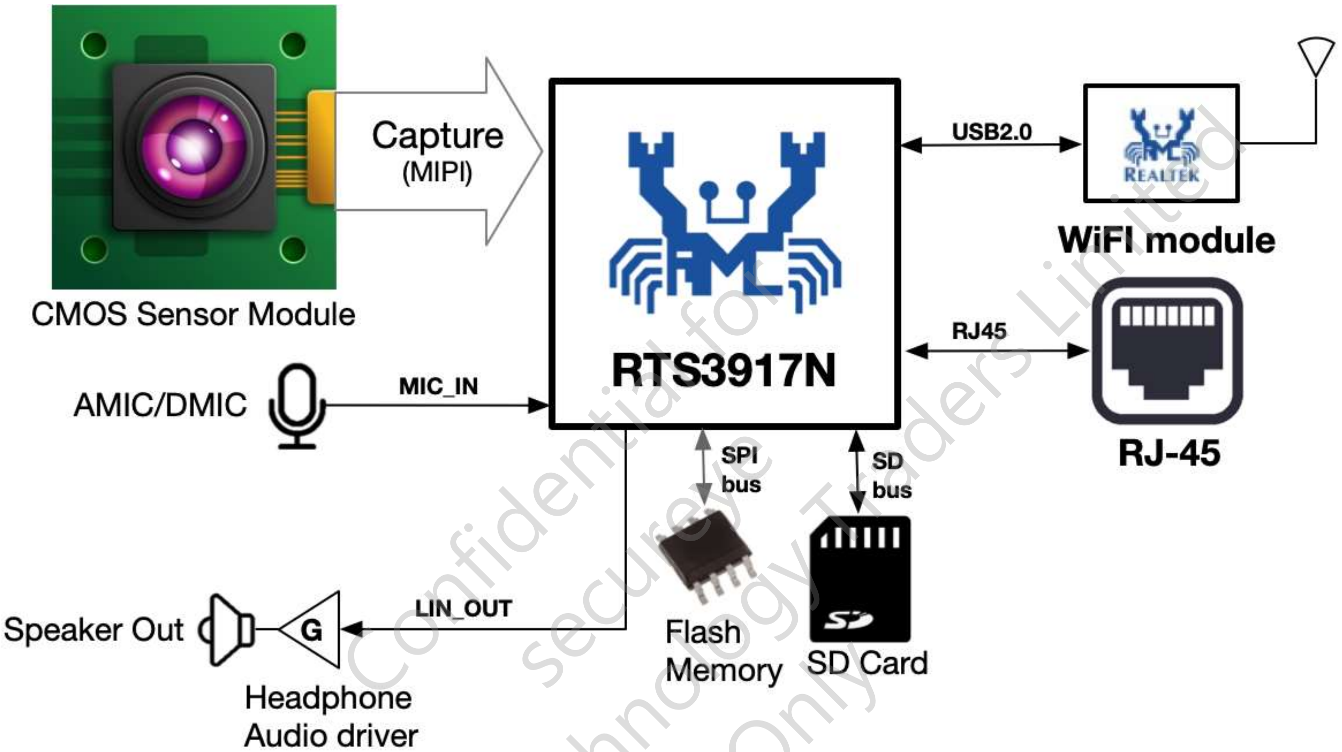
Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide. In that event, please contact your Realtek representative for additional information that may help in the development process.

REVISION HISTORY

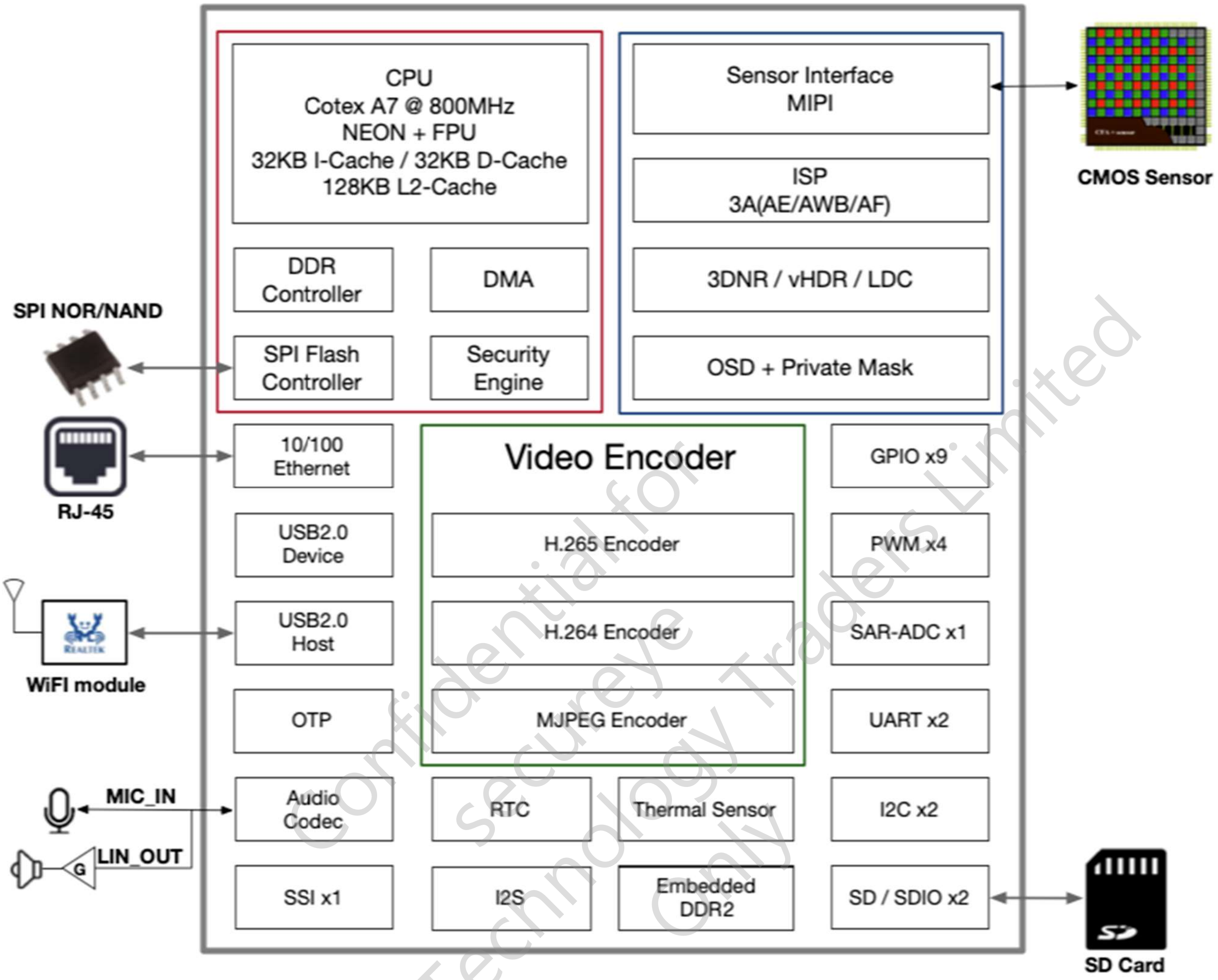
Revision	Release Date	Summary
1.0	2025/10/21	First release

Overview

As a professional system-on-chip (SoC), the RTS3917N is designed for the IP camera. Based on application requirements, it encodes multiple streams in H.265/H.264 format and provides superior image signal processing (ISP) performance, high encoded video quality, and a high-performance intelligent acceleration engine. With these features, it meets customers' requirements for product functions, performance, and image quality, and helps customers significantly reduce the engineering bill of materials costs. The following describes typical 3M IP camera solutions.



Architecture Diagram



Key Features

Central Processing Unit

- ARM Cortex-A7
- CPU Clock Frequency 800MHz
- Neon and FPU
- 128KB L2 Cache/32KB I-Cache/ 32KB D-Cache

Integrated Memory

- Support 16-bit DDR2
- Integrated 512Mbits DDR2

SPI NOR Flash

- Support single/dual/quad channel read/write mode
- Support SPI NOR flash
- Support 4 byte address mode for 32MB flash

SPI NAND Flash

- Support single/dual/quad channel read/write mode
- Support up to 2Gbits SPI NAND flash with 2KB page size

Security Engine

- Supports AES-128/192/256-ECB/ CBC/CTR/GCM/CCM
- Supports RSA-512/1024/2048/3072
- Supports SHA256/HMAC-SHA256
- Supports TRNG
- AES data block length 128bits
- Support multi-buffer encryption/decryption
- Support inter-buffer data splicing
- Support auto-padding

- OTP (One Time Programmable) 16kbits

Ethernet

- Embedded 10/100 Mac and Physical Layer

Wireless

- Support USB2.0 Wi-Fi module
- Support SDIO Wi-Fi module

Sensor interface

- MIPI 2-lane interface up to 12-bit Raw data
- HCLK support 24MHz/27MHz or 54MHz/37.125MHz or 74.25MHz

Image Signal Processor

- Support individual BLC (Black Level Cancellation) or OBC (Optical Black Cancellation)
- Individual LSC (Lens Shading Correction) for Gr/R/B/Gb channels
- LSC supports both of circle and grid compensation
- On-the-fly DPC (defect pixel compensation)
- Support individual 3A (AE/AWB/AF) statistics
- Advanced color interpolation (de-mosaicking) with false color reduction
- Programmable R/G/B Gamma
- Support WDR engine and auto WDR control
- 2 exposures vHDR (Video HDR)
- Support Auto Dynamic Range Compression
- Support de-haze engine

- Flexible CSC (Color Space Conversion) with default of BT.709 standard
- Advance EEH (Edge Enhance) with improvement of detail and sharpness
- Capability of horizontal LDC (Lens Distortion Correction)
- Whole new designed 3DNR (3D Noise Reduction) with suppression of noise vibration
- Support private mask
- Support zoom out with 5 channels
- OSD supports 6 areas and monochrome pictures with 5 channels
- OSD supports alpha-blending process
- Support ISP tuning tools for the PC through Ethernet/Wi-Fi

H.265/H.264 Encoder

- Support H.265/H.264 encoder
- Support H.265/H.264 MBRC
- Support H.265/H.264 CBR/VBR/CVBR Rate control mode
- Support H.265/H.264 ROI Map
- Support video rotates
- Maximum H.265/H.264 resolution is 3M
- H.264 encoder support Base Line/Main Profile /High Profile
- H.265 encoder support Main Profile
- Supports up to quarter-pixel
- Supports frame level and MB level rate control
- Supports ROI encoding with custom QP map

JPEG Encoder

- Supports JPEG baseline encoding
- Supports YUV422 or YUV420 formats
- Supports max. 3M with 25fps encoding

Video Streaming

- Support H.265/H.264 2-stream : 3M@25fps + VGA@25fps
- H.265/H.264 stream support CBR/VBR/CVBR
- Support MJPEG 3M@25fps

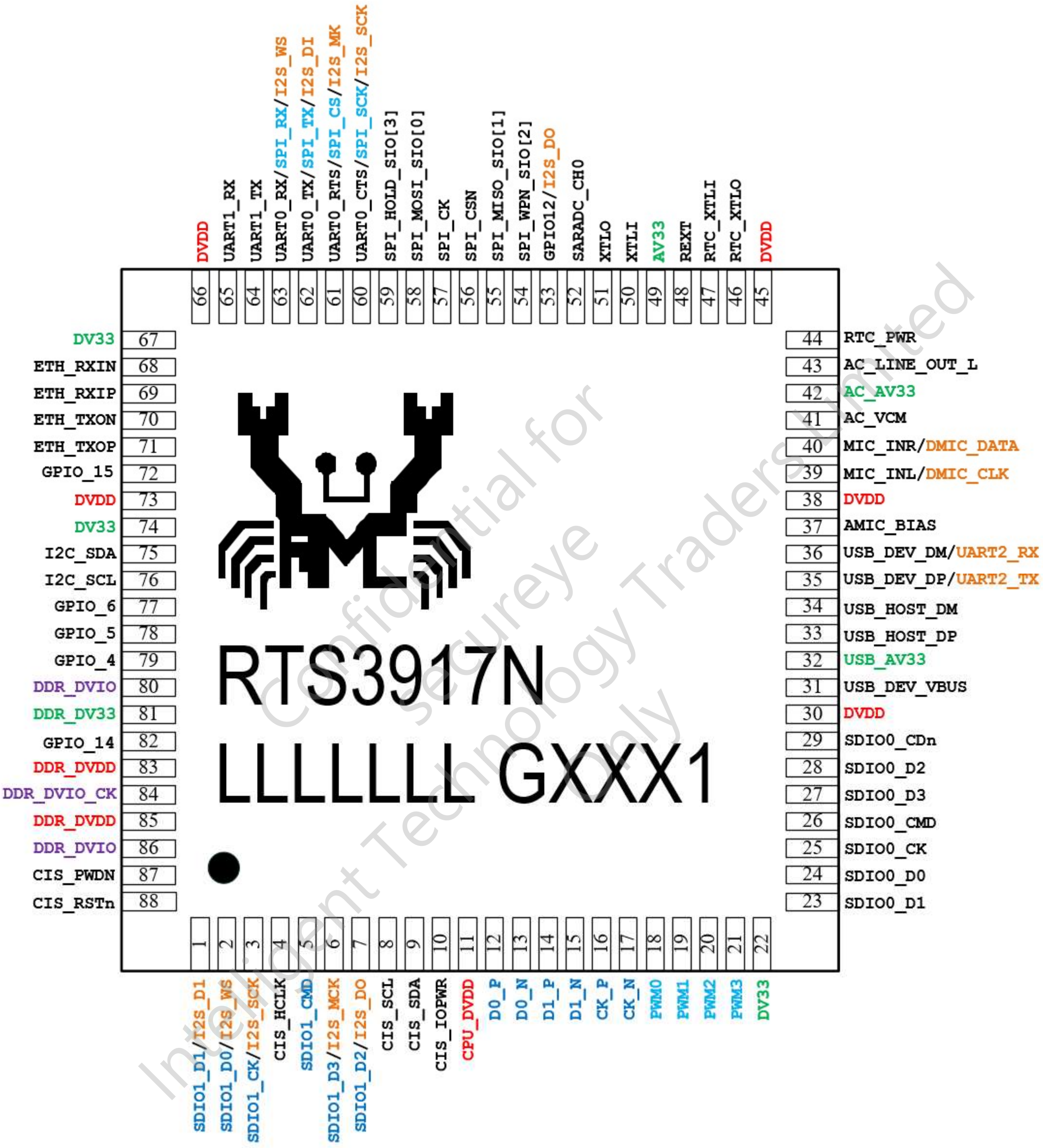
Audio Processor

- Build-in one 24-bit audio codec
- Support mono ADC/DAC
- Support DMIC input
- Support I2S
- Support sample rate: 192kHz/ 176.4kHz/ 96kHz/ 88.2kHz/ 48kHz/ 44.1kHz/ 32kHz/ 24 kHz/ 22.05KHz/ 16 kHz/ 11.025 KHZ/ 8kHz
- Support G.711/G.726/AAC encoding
- Software Acoustic echo cancellation

Peripheral

- GPIO x9, PWM x4, SARADC 1-channel
- UART x2
- I2C interface, SSI interface
- USB Host 2.0 & USB Device 2.0
- SD 2.0 x2 or SD 2.0 x1 & eMMC x1
- RTC
- Embedded thermal sensor

Pin Assignment



Note: Green package is indicated by a 'G' in the location marked 'T'. The silicon version number is shown in the location marked 'V'.

Appendix for STQC submission

To whom it may have concern:

We hereby confirm, based on our design, best of our knowledge, and belief, that the SoC model RTS3917N do not include the following feature:

- TEE (Trusted Execution Environment)
- Intellectual property protection technologies

This appendix is attached to ensure transparency and clarity regarding its capabilities and development scope. It is intended solely for submission to STQC as proof for clearing the SoC scope, and is not for general distribution.

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